Fully penetrant within whose lifetime? Step-by-step calculations

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October 22, 2026

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This document reproduces the calculations behind the paper *Fully Penetrant Within Whose Lifetime? Background Mortality and the Penetrance of Huntington's Disease Alleles*. Each section follows one step of Supplementary Methods S1, uses carriers of 40 CAG repeats in Mexico as the worked example, and checks the result against the outputs of the analysis scripts in `code/`. All code is base R; `knitr` is used only to print tables.

Data

The main analysis uses the 2019 World Health Organization (WHO) life tables for both sexes: age-specific death rates ${}_nm_x$ (indicator `LIFE_0000000029`) and life expectancies e_x (indicator `LIFE_0000000035`) for 14 countries. ${}_nm_x$ is the death rate in the age group that starts at age x and spans n years.

```
lt <- read.csv("code/data/who_life_tables.csv", stringsAsFactors = FALSE)
lt <- lt[order(lt$country, lt$year, lt$age_start), ]
table(lt$country, lt$year)
```

```
      2019 2023
```

```
  ARG   19    0
  BRA   19    0
  CAN   19    0
  CHL   19    0
  COL   19    0
  ESP   19   19
  GBR   19   19
  GTM   19    0
  HND   19    0
  MEX   19   19
  PER   19    0
  SLV   19    0
  SWE   19    0
  USA   19    0
```

```
knitr::kable(lt[lt$country == "MEX" & lt$year == 2019, ], row.names = FALSE,
             caption = "WHO 2019 life table for Mexico, both sexes")
```

Table 1: WHO 2019 life table for Mexico, both sexes

country	year	age_group	age_start	nmx	ex
MEX	2019	00-01	0	0.013837	75.01791
MEX	2019	01-04	1	0.000316	75.05598
MEX	2019	05-09	5	0.000251	71.14889
MEX	2019	10-14	10	0.000300	66.23519
MEX	2019	15-19	15	0.000799	61.33093
MEX	2019	20-24	20	0.001375	56.56654
MEX	2019	25-29	25	0.001689	51.93950
MEX	2019	30-34	30	0.001907	47.35879
MEX	2019	35-39	35	0.002350	42.78853
MEX	2019	40-44	40	0.003110	38.26477
MEX	2019	45-49	45	0.004276	33.82521
MEX	2019	50-54	50	0.005925	29.50218
MEX	2019	55-59	55	0.008813	25.31414
MEX	2019	60-64	60	0.013243	21.34212
MEX	2019	65-69	65	0.019007	17.63246
MEX	2019	70-74	70	0.028214	14.14230
MEX	2019	75-79	75	0.043411	10.90930
MEX	2019	80-84	80	0.071874	7.95678
MEX	2019	85PLUS	85	0.173063	5.34734

Step 1: distribution of age at onset

Following Langbehn et al. (2004), age at onset T for a carrier of c CAG repeats follows a logistic distribution with mean $\mu_c = 21.54 + \exp(9.556 - 0.1460c)$ and variance $\sigma_c^2 = 35.55 + \exp(17.72 - 0.3269c)$. The logistic scale parameter is $s_c = \sigma_c\sqrt{3}/\pi$.

```
langbehn <- function(cag, sd_scale = 1) {
  mu <- 21.54 + exp(9.556 - 0.1460 * cag)
  sigma <- sqrt(35.55 + exp(17.72 - 0.3269 * cag)) * sd_scale
  s <- sigma * sqrt(3) / pi
  list(mu = mu, sigma = sigma, s = s,
       cdf = function(t) plogis(t, location = mu, scale = s),
```

```
       pdf = function(t) dlogis(t, location = mu, scale = s))
}
m40 <- langbehn(40)
c(mu = m40$mu, sigma = m40$sigma, scale = m40$s)
```

```
      mu    sigma    scale
62.63967 11.81141  6.51197
```

Step 2: net penetrance among carriers alive at age 18

Net penetrance is the conditional probability of onset by age t given no onset by $a_0 = 18$:

$$N_c(t) = P(T \le t \mid T > a_0) = \frac{F_c(t) - F_c(a_0)}{1 - F_c(a_0)},$$

and its derivative, $g_c(t) = f_c(t)/[1 - F_c(a_0)]$, is the density of onset age among those carriers.

```
a0 <- 18
net_pen <- function(cag, t, a0 = 18, sd_scale = 1) {
  m <- langbehn(cag, sd_scale)
  (m$cdf(t) - m$cdf(a0)) / (1 - m$cdf(a0))
}
c(F40_at_18 = m40$cdf(18), net_by_65 = net_pen(40, 65), net_by_75 = net_pen(40, 75),
  net_by_85 = net_pen(40, 85))
```

```
 F40_at_18  net_by_65  net_by_75  net_by_85
0.00105304 0.58920340 0.86953830 0.96870920
```

```
## The conditional density integrates to 1 over [18, infinity)
integrate(function(t) m40$pdf(t) / (1 - m40$cdf(a0)), a0, Inf)$value
```

```
[1] 1
```

Step 3: background mortality as a piecewise-constant hazard

Each WHO rate is treated as a constant hazard within its age group, and the rate for ages 85 and older applies from 85 onward: $h_k(t) = {}_nm_x$ for $x \le t < x + n$.

```
breaks <- c(0, 1, seq(5, 85, 5))
make_hazard <- function(cc, yr = 2019, open = "nmx") {
  d <- lt[lt$country == cc & lt$year == yr, ]
  h <- d$nmx
  if (open == "inv_e85") h[length(h)] <- 1 / d$ex[nrow(d)]
  function(t) h[findInterval(t, breaks)]
}
h_mex <- make_hazard("MEX")
c(age_22 = h_mex(22), age_62 = h_mex(62), age_82 = h_mex(82))
```

```
   age_22    age_62    age_82
0.0013749 0.0132428 0.0718740
```

Step 4: survival from age 18

$$S_k(t \mid a_0) = \exp\left\{-\int_{a_0}^{t} h_k(v)\,dv\right\} = \exp\left\{-\sum_j h_{kj}\,\Delta_j(t)\right\},$$

where $\Delta_j(t)$ is the number of years of age group j between a_0 and t. The sum is exact for a piecewise-constant hazard.

```
make_surv <- function(haz) {
  function(t, a0) {
    sapply(t, function(tt) {
      if (tt <= a0) return(1)
      knots <- sort(unique(c(a0, breaks[breaks > a0 & breaks < tt], tt)))
      exp(-sum(haz(knots[-length(knots)]) * diff(knots)))
    })
  }
}
S_mex <- make_surv(h_mex)
S_esp <- make_surv(make_hazard("ESP"))
```

The table below writes out the sum for Mexico term by term (Supplementary Table S2).

```
d <- lt[lt$country == "MEX" & lt$year == 2019 & lt$age_start >= 15 & lt$age_start < 80, ]
years_after_18 <- pmin(d$age_start + 5, 80) - pmax(d$age_start, 18)
contribution <- d$nmx * years_after_18
H <- cumsum(contribution)
tabS2 <- data.frame(age_group = d$age_group, years_after_18, nmx = d$nmx,
                    contribution, cumulative_hazard = H, to_age = d$age_start + 5,
                    survival_pct = 100 * exp(-H))
knitr::kable(tabS2, digits = c(0, 0, 6, 6, 6, 0, 1), row.names = FALSE,
             caption = "Survival from age 18 in Mexico, 2019, computed term by term")
```

Table 2: Survival from age 18 in Mexico, 2019, computed term by term

age_group	years_after_18	nmx	contribution	cumulative_hazard	to_age	survival_pct
15-19	2	0.000799	0.001599	0.001599	20	99.8
20-24	5	0.001375	0.006875	0.008473	25	99.2
25-29	5	0.001689	0.008445	0.016918	30	98.3
30-34	5	0.001907	0.009534	0.026453	35	97.4
35-39	5	0.002350	0.011751	0.038204	40	96.3
40-44	5	0.003110	0.015548	0.053752	45	94.8
45-49	5	0.004276	0.021380	0.075132	50	92.8
50-54	5	0.005925	0.029625	0.104757	55	90.1
55-59	5	0.008813	0.044066	0.148823	60	86.2
60-64	5	0.013243	0.066214	0.215037	65	80.7
65-69	5	0.019007	0.095034	0.310071	70	73.3
70-74	5	0.028214	0.141069	0.451140	75	63.7
75-79	5	0.043411	0.217055	0.668195	80	51.3

For example, from 18 to 60 the interval covers 2 years of $[15, 20)$ and 5 years of each group from $[20, 25)$ to $[55, 60)$:

```
H60 <- 2 * d$nmx[d$age_start == 15] + 5 * sum(d$nmx[d$age_start >= 20 & d$age_start <= 55])
H80 <- H60 + 5 * sum(d$nmx[d$age_start >= 60 & d$age_start <= 75])
c(H60 = H60, S60 = exp(-H60), S60_function = S_mex(60, 18),
  H80 = H80, S80 = exp(-H80), S80_function = S_mex(80, 18))
```

```
       H60         S60 S60_function         H80         S80 S80_function
  0.148823    0.861721    0.861721    0.668195    0.512633    0.512633
```

```r
c(Spain_S60 = S_esp(60, 18), Spain_S80 = S_esp(80, 18))
```

```
Spain_S60 Spain_S80
 0.946702  0.712905
```

Step 5: crude penetrance

Let D be age at death from other causes. Assuming T and D are independent, conditioning on onset at age u gives $P(D > u \mid D > a_0) = S_k(u \mid a_0)$, so

$$C_{ck}(t) = \int_{a_0}^{t} S_k(u \mid a_0)\, g_c(u)\, du = \frac{1}{1 - F_c(a_0)} \int_{a_0}^{t} f_c(u)\, S_k(u \mid a_0)\, du.$$

Crude lifetime penetrance is $C_{ck}(\infty)$, computed with an upper limit of 130 years.

```r
crude_pen <- function(cag, cc, a0 = 18, yr = 2019, open = "nmx", sd_scale = 1, upper = 130) {
  m <- langbehn(cag, sd_scale)
  S <- make_surv(make_hazard(cc, yr, open))
  integrate(function(u) m$pdf(u) * S(u, a0), a0, upper, subdivisions = 2000)$value /
    (1 - m$cdf(a0))
}
c(Mexico = crude_pen(40, "MEX"), Spain = crude_pen(40, "ESP"),
  gap_points = 100 * (crude_pen(40, "ESP") - crude_pen(40, "MEX")))
```

```
    Mexico      Spain gap_points
  0.786720   0.890364  10.364342
```

Setting $S_k = 1$ recovers net penetrance, so crude penetrance can never exceed net penetrance:

```r
no_death <- integrate(function(u) m40$pdf(u), a0, 130)$value / (1 - m40$cdf(a0))
c(crude_without_death = no_death, net_by_130 = net_pen(40, 130))
```

```
crude_without_death          net_by_130
           0.999968            0.999968
```

As a numerical check, a Riemann sum on a grid of 0.001 years gives the same answer:

```r
grid_crude <- function(k, cc, a0 = 18, dt = 0.001) {
  m <- langbehn(k); h <- make_hazard(cc)
  u <- seq(a0, 130, by = dt)
  S <- exp(-cumsum(c(0, h(u[-length(u)]) * dt)))
  sum(m$pdf(u) * S) * dt / (1 - m$cdf(a0))
}
100 * c(Mexico = grid_crude(40, "MEX") - crude_pen(40, "MEX"),
        Spain = grid_crude(40, "ESP") - crude_pen(40, "ESP"))
```

```
     Mexico       Spain
-0.00301198 -0.00206903
```

Step 6: median age at death and the labeled points in Figure 1

```r
med_death <- function(cc) uniroot(function(t) make_surv(make_hazard(cc))(t, 18) - 0.5,
                                  c(19, 129))$root
dot <- function(cc) {
  m <- med_death(cc); S <- make_surv(make_hazard(cc))
  integrate(function(u) m40$pdf(u) * S(u, 18), 18, m)$value / (1 - m40$cdf(18))
```

```
}
fig_cc <- c("ESP", "GBR", "BRA", "MEX", "GTM")
knitr::kable(data.frame(country = fig_cc,
                        median_age_at_death = sapply(fig_cc, med_death),
                        crude_by_median_pct = 100 * sapply(fig_cc, dot)),
             digits = 1, row.names = FALSE)
```

country	median_age_at_death	crude_by_median_pct
ESP	85.9	88.3
GBR	84.8	86.1
BRA	80.9	78.4
MEX	80.3	76.8
GTM	77.7	72.1

Table 1: net and crude penetrance by repeat length and country

```
countries <- c("ESP", "SWE", "GBR", "CAN", "USA", "CHL", "PER", "COL", "ARG", "BRA",
               "MEX", "SLV", "HND", "GTM")
cags <- 40:45
tab1 <- t(sapply(cags, function(k) sapply(countries, function(cc) crude_pen(k, cc))))
rownames(tab1) <- cags
knitr::kable(100 * t(tab1), digits = 1,
             caption = "Crude lifetime penetrance (%) by country (rows) and CAG repeat length (columns)")
```

Table 3: Crude lifetime penetrance (%) by country (rows) and CAG repeat length (columns)

	40	41	42	43	44	45
ESP	89.0	93.7	96.2	97.6	98.4	98.8
SWE	89.1	93.8	96.2	97.5	98.1	98.6
GBR	86.9	92.1	95.0	96.6	97.6	98.2
CAN	87.6	92.5	95.1	96.6	97.5	98.1
USA	83.0	88.9	92.4	94.5	95.9	96.8
CHL	86.3	91.6	94.5	96.2	97.2	97.8
PER	84.1	89.6	92.9	94.8	96.0	96.8
COL	84.6	90.1	93.1	94.8	95.9	96.6
ARG	81.6	88.4	92.4	94.7	96.1	97.0
BRA	80.1	86.7	90.6	93.1	94.6	95.7
MEX	78.7	85.5	89.8	92.5	94.2	95.4
SLV	75.8	82.7	87.1	90.1	92.2	93.7
HND	75.4	83.3	88.3	91.5	93.5	94.8
GTM	74.9	82.6	87.5	90.7	92.8	94.3

```
knitr::kable(data.frame(cag = cags, median_onset = sapply(cags, function(k) langbehn(k)$mu),
                        net_by_75_pct = 100 * sapply(cags, function(k) net_pen(k, 75))),
             digits = 1, row.names = FALSE, caption = "Onset model by repeat length")
```

Table 4: Onset model by repeat length

cag	median_onset	net_by_75_pct
40	62.6	87.0
41	57.1	95.7
42	52.2	98.7
43	48.1	99.7
44	44.5	99.9
45	41.3	100.0

Check against the script output (`code/output/table1.csv`):

```
saved <- read.csv("code/output/table1.csv")
max(abs(as.matrix(saved[, countries]) - tab1))
```

```
[1] 1.88738e-15
```

Crude penetrance at 40 repeats against remaining life expectancy at age 40:

```
e40 <- sapply(countries, function(cc) lt$ex[lt$country == cc & lt$year == 2019 & lt$age_start == 40])
knitr::kable(data.frame(country = countries, e40 = e40, crude40_pct = 100 * tab1["40", ])[order(e40), ]
             digits = 1, row.names = FALSE)
```

country	e40	crude40_pct
GTM	35.7	74.9
HND	35.9	75.4
SLV	37.3	75.8
MEX	38.3	78.7
BRA	38.8	80.1
ARG	39.1	81.6
USA	40.7	83.0
COL	40.8	84.6
PER	41.6	84.1
GBR	42.3	86.9
CHL	42.9	86.3
CAN	43.4	87.6
SWE	43.5	89.1
ESP	43.9	89.0

Sensitivity analyses (Supplementary Table S1)

```
sens <- function(f) sapply(c("MEX", "ESP", "GBR"), function(cc) f(40, cc))
S1 <- rbind(
  main               = sens(function(k, cc) crude_pen(k, cc)),
  hazard_85_inv_e85  = sens(function(k, cc) crude_pen(k, cc, open = "inv_e85")),
  life_tables_2023   = sens(function(k, cc) crude_pen(k, cc, yr = 2023)),
  start_age_30       = sens(function(k, cc) crude_pen(k, cc, a0 = 30)),
  onset_sd_x0.8      = sens(function(k, cc) crude_pen(k, cc, sd_scale = 0.8)),
  onset_sd_x1.2      = sens(function(k, cc) crude_pen(k, cc, sd_scale = 1.2)))
S1 <- cbind(100 * S1, gap_ESP_MEX_points = 100 * (S1[, "ESP"] - S1[, "MEX"]))
knitr::kable(S1, digits = 1, caption = "Crude lifetime penetrance (%) at 40 repeats")
```

Table 5: Crude lifetime penetrance (%) at 40 repeats

	MEX	ESP	GBR	gap_ESP_MEX_points
main	78.7	89.0	86.9	10.4
hazard_85_inv_e85	78.7	89.0	86.9	10.3
life_tables_2023	79.0	89.3	86.3	10.3
start_age_30	79.9	89.3	87.3	9.4
onset_sd_x0.8	80.2	90.6	88.6	10.3
onset_sd_x1.2	77.0	87.3	85.1	10.3

Global supplementary analysis (Supplementary Figure S1)

The global analysis repeats Step 5 for 201 countries with single-year death rates from the United Nations World Population Prospects 2024 (`code/extract_wpp2024.R`, `code/global_penetrance.R`). The results are read from the saved output here because the full run takes a few seconds per country.

```
g <- read.csv("code/output/global_penetrance.csv")
nrow(g)
```

```
[1] 201
```

```
summary(100 * g$crude40)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   22.4    70.3    77.8    76.7    83.9    91.1
```

```
g$rank <- rank(g$crude40)
knitr::kable(g[g$name %in% c("Mexico", "Guatemala", "Spain", "Sweden", "Canada", "Japan",
                             "China", "India", "United States of America"),
               c("name", "e40", "crude40", "rank")],
             digits = 3, row.names = FALSE)
```

name	e40	crude40	rank
India	35.142	0.745	77
Guatemala	35.869	0.752	85
Mexico	37.885	0.777	100
United States of America	40.959	0.828	142
China	39.671	0.838	150
Canada	43.651	0.877	177
Spain	44.209	0.893	188
Sweden	43.921	0.895	191
Japan	45.225	0.900	194

```
knitr::kable(read.csv("code/output/global_vs_who.csv"), digits = 3,
             caption = "Crude penetrance at 40 repeats: UN WPP 2024 against WHO, 14 main countries")
```

Table 6: Crude penetrance at 40 repeats: UN WPP 2024 against WHO, 14 main countries

iso3	wpp	who	diff_pts
ESP	0.893	0.890	0.279
SWE	0.895	0.891	0.353

iso3	wpp	who	diff_pts
GBR	0.870	0.869	0.118
CAN	0.877	0.876	0.125
USA	0.828	0.830	-0.172
CHL	0.852	0.863	-1.169
PER	0.808	0.841	-3.213
COL	0.827	0.846	-1.870
ARG	0.814	0.816	-0.192
BRA	0.803	0.801	0.228
MEX	0.777	0.787	-0.944
SLV	0.732	0.758	-2.605
HND	0.756	0.754	0.229
GTM	0.752	0.749	0.339

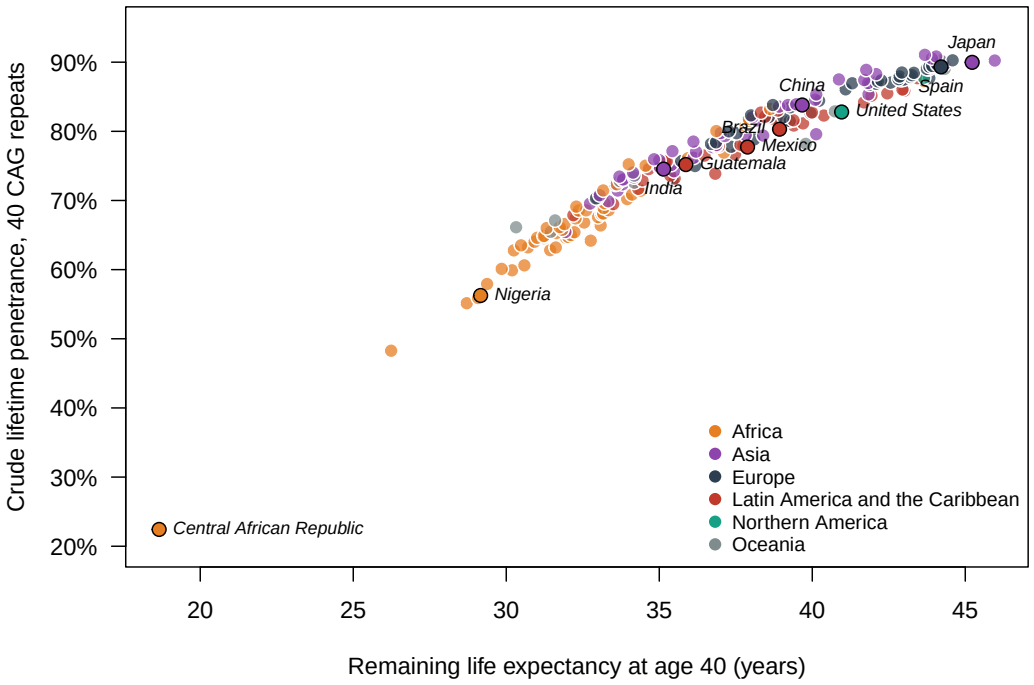
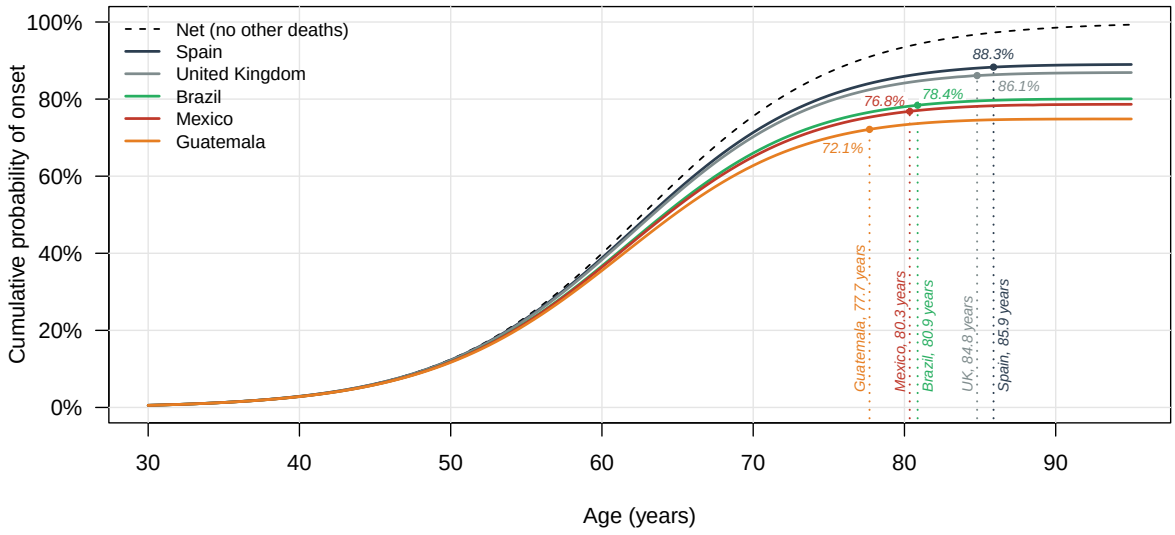


Figure 1



Session information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: aarch64-apple-darwin20
Running under: macOS 27.0.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK versi

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
 [1] compiler_4.5.1    fastmap_1.2.0     cli_3.6.5         tools_4.5.1
 [5] htmltools_0.5.8.1 rstudioapi_0.17.1 yaml_2.3.10       rmarkdown_2.30
 [9] knitr_1.50        xfun_0.60         digest_0.6.38     rlang_1.3.0
[13] evaluate_1.0.5
```